Recovering Versioned Releases Across Build and Publish Boundaries

***Abstract*—We analyze three release anomalies: availability inversions (consumers start before required artifacts are published), orphaned releases (provider success erases unfinished consumer intent), and retry version drift (changed inherited channels alter targets). We present a saga-inspired protocol combining deterministic planning, stage-aware scheduling, and consumer-owned release obligations. Git history and records form the ledger without another persistent release-state store. Consumer windows retain undelivered intent; train and fresh-work windows distinguish prerelease targets from completed work. Under an explicit retry invariant, the single-history model establishes eight conditional guarantees. All 120 controlled fault episodes recover after repair; an analytical source-discharge alternative loses work in 84 (70%). This compares admission rules, not competing tools or production failure rates. A skill supports commit release-intent diagnostics, plan inspection, and CI releases. Proposed fleet and worker extensions preserve source-owned records and reuse verified provider outputs. The experiments report will include Linux and LLVM distributed-build scenarios: pessimistic runtimes above $1\times$ and optimistic runtimes below $1\times$ each original pipeline's elapsed time. These scenarios are prospective. Full implementation refinement and external exactly-once effects remain outside the claims.**

***Index Terms*—software release engineering, continuous integration and delivery, dependency scheduling, recovery, idempotence**

I. Introduction

Consider an npm asset library, its dependent UI package, a Docker image packaging that UI, and a second image built from the first. The UI consumes local output; an isolated registry-based image builder needs its base published. A global build-then-publish schedule fails, while serializing every package makes local-artifact consumers wait for unnecessary uploads.

If the final image fails after its providers publish, retry should retain its release and target version. The engine must distinguish which versions are owed, which stages may run, and which obligations survive failure.

Release engineering spans more than compilation [1]. Without an atomic commit across registries, execution resembles a saga [2]: steps commit separately. Our protocol uses forward recovery because published artifacts may not be retractable.

Existing tools schedule stages, compute versions, and coordinate publication [3], [4], [5], [6], [7]. We study their composition with a ledger that retains failed consumers' obligations after providers succeed. Consumer-based admission keeps propagated intent pending according to the consumer's own release window. Separate windows distinguish prerelease-train intent from fresh, undelivered work.

We present ReleaseWeave and its release-intent language RIL, using pseudonyms for anonymous review; campaign records retain exact artifact identities. Git history and tags form the ledger; ordinary commands build and publish. No additional persistent release-state database is required, apart from storage used by registries, build tools, and applications.

A package is a versioned unit of software, specifications, or documentation, using common intent, dependency, command, and completion interfaces. Guarantees require conforming integrations. The graph gives provisioned CI a stable entry point for affected tests and releases; package configuration still changes, and new credentials or runners may require workflow changes.

The paper makes three contributions:

- An ecosystem-independent, saga-inspired release protocol combining local-artifact and published-artifact dependencies with explicit release intent and per-package completion records, including a description of fleet composition across repository-owned histories (Section II).
- A self-contained model with deterministic planning and eight recovery, discharge, and scope/channel guarantees under stated assumptions (Sections III–IV). Counterexamples explain why consumer windows and stable retry predicates matter.
- An evaluation of a released binary using differential planning, targeted vectors, injected failures, historical replay, and scale measurements (Section V), with the model-to-implementation gap made explicit.

The mathematical contribution is the admission and discharge discipline, not a new topological-sorting or Git-storage primitive.

II. System and Release Protocol

A. A Release Crossing Two Availability Boundaries

The running example is npm `assets` $\rightarrow$ npm `ui` $\rightarrow$ Docker `base` $\rightarrow$ Docker `app`. Figure 1 uses provider-to-consumer arrows. The app builder obtains its base through a registry, without a shared local image store.

Availability modes describe integration choices, not ecosystems. Here, `ui` and `base` use workspace output; `app` pulls its base from a registry. The provider setting `isBuildWaitingPublish: true` therefore belongs on `base`. Both npm providers keep the default local-build mode. Other integrations may require npm publication or share local Docker images. Consumer publication follows provider publication in both modes.

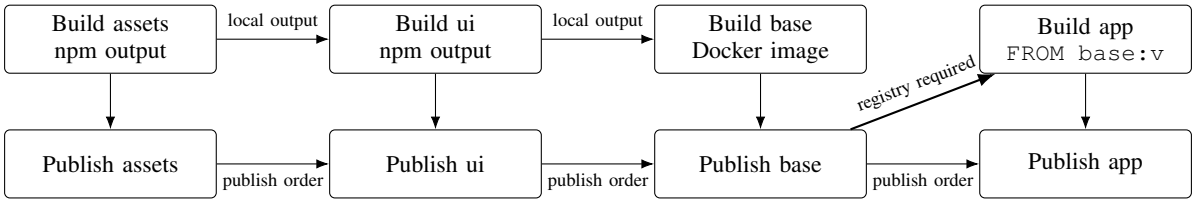


Fig. 1. Stage dependencies, not a measured timeline. Publishing `assets` may overlap building `ui`, whose publication may overlap building `base`. Building `app` must wait for `base`'s publication. Availability modes are configured requirements: an npm dependency fetched from a registry would also require publication.

Suppose `assets` is at 1.4.2, `ui` at 3.2.0, `base` at 0.8.2, and `app` at 2.1.0. An intent to release an assets feature and patch all transitive consumers plans versions 1.5.0, 3.2.1, 0.8.3, and 2.1.1. After the first three packages publish and are recorded, `app` fails. A planner that only propagates from *currently unreleased providers* can now lose `app`, leaving an orphan in the release ledger. Our planner instead asks whether `app` has received that intent. Its own pending window is unchanged, so a retry still owes version 2.1.1.

B. From Repository Inputs to a Plan

The planner reads a fixed revision, package and dependency definitions, messages, records, commands and availability requirements. It selects packages, versions and reasons before publication; changed semantic inputs require replanning. The extended protocol additionally fixes repository-qualified histories, consumer boundaries and verified withdrawal inventory/receipts. Execution placement is separate from this semantic input.

Release intent extends Conventional Commits [8]. For example, `feat(assets)^^: add theme assets` directly requests a minor release of `assets` and propagates patch intent to transitive consumers. An unmarked unit affects its own scope only. A scope can explicitly name packages or be derived from changed paths. A commit can contain multiple independent units, each retaining its own source set and propagation depth. The formal model consumes normalized units, not a particular parser syntax. The complete language grammar is outside this paper [9]. A diagnostics command validates literal intent syntax and parser policy without committing; it does not verify scope or plan correctness.

Release-worthy descriptions can form version-specific notes, including for private projects without changelogs. Humans can adapt them for social drafts or management reports; this unmeasured use does not automate posting. Notes are narratives; tags discharge release obligations.

Version bumps and prerelease channels are separate axes; a channel transition need not request a bump. Cancellation, correction, holds, and exact versions govern pending intent without unpublishing artifacts. These controls motivate the model predicates; their CLI forms are outside our contribution.

C. Execution, Recording, and Recovery

The executor schedules version reconciliation, build and publish tasks behind provider availability gates, with separate build/publish limits. Independent ready tasks may overlap; deterministic planning does not fix completion order. Publication creates a recording obligation: only the durable release record discharges work. The model combines those steps; implementations must report recording and finalization failures separately. A retry reconstructs pending work from records, without an additional persistent recovery database.

A known publication failure leaves its obligation pending. An absent tag, however, cannot distinguish failure from successful-but-unrecorded publication. Before recording, reconcile artifact identity or safely repeat the publisher; forward recovery alone does not resolve uncertainty [10]. Locks coordinate participating writers, not arbitrary registry transactions.

a) Optional worker execution.: Execution nodes are distinct from repository peers and apply in every history mode. The CI-entry orchestrator acquires all participant locks in stable order, revalidates membership, and fixes the plan before dispatch. Explicit workers cannot initiate releases. Placement and completion order do not change semantic inputs or selected versions; node capacity intersects run-wide stage budgets. Shared preparation precedes isolated task worktrees. Temporary Git branches carry output bytes or retrievable immutable bundles, never release intent or completion records. Consumers verify exact input/output identities and required bytes without rebuilding provider closures. Tests and recording gates remain. The orchestrator alone admits results and records releases; publication and recording serialize per owner even when builds overlap. A timed-out publisher requires acknowledged cancellation, effective fencing or reconciliation before safe reassignment; an ownership-generation field alone does not fence effects. Transport loss cannot erase release obligations. This profile is specified, not evaluated by the campaigns below.

D. Fleet Releases Across Repository Histories

A *control repository* holds deployment configuration—manifests, environment settings and artifact selections—separately from application source, following Puppet and GitOps practice [11], [12]. Here it composes releases; source repositories retain completion records. Configuration commits do not prove completion.

History modes [9] are single history (one package, monorepo or control-history wrapper), specified sources matching control pins, and discovered peers with their own

configurations, observed heads and advisory link pins. Peer links form a tree; a star grants its centre no policy authority.

a) Ownership and admission.: Commit identities are repository-qualified full object IDs; ancestry remains local. For source repository r with fixed head h_r, let $A_r(x) = reach_r(h_r) \setminus reach_r(x)$. If consumer D's stable and newest releases prove boundaries $s_{D,r}$ and $b_{D,r}$ in r, its train and fresh windows are

$$W(D;r) = A_r(s_{D,r}), \qquad W^{\circ}(D;r) = A_r(s_{D,r}) \cap A_r(b_{D,r}).$$

Source units use the consumer's window in that source history. For an absent baseline, its subtracted reach is empty; missing evidence for an existing tag requires an explicit association, not an invented boundary. Automatic evidence ties the exact consumer tag to a control checkpoint or to its release commit's recorded link route. Tag-only releases remain valid but may need explicit baseline tuples. Incomparable directives requiring one winner are refused; timestamps and output sorting cannot create semantic precedence. These window definitions describe the extension, not a multi-history proof of Section IV.

b) Partial fleet release.: Before publication, revalidate the package's provider and shared-group input closure. Group membership can introduce cycles into that closure computation; this does not permit cycles in the publication DAG. Only exact native record and settlement heads advance the expected snapshot. Completed worker output also needs unchanged effective inputs; an admitted head alone is insufficient. Peer-link settlement precedes publication when release commits provide boundary evidence; tag-only operation must not force a settlement commit. Recording routes take owner lanes in stable order. Source records become durable before pushed pointers or control checkpoints refer to them. A failed checkpoint preserves source success and requires explicit repair, not republication. Recording failures block dependent work; cleanup cannot rewrite successes or release exclusion while unfenced effects remain active. These mechanisms are forward recovery across separate ledgers, not an atomic fleet transaction.

E. Integration Limits

A Go module consumer needs published tagged source, not upstream compiled output; local `go.work` or `replace` can bind a checkout [13]. This configuration-specific example is unevaluated.

Conforming integrations supply stable identities, dependencies, availability, trustworthy stage completion and durable records. Uncertain publication requires reconciliation or safe repetition. Specifications can use validation/rendering and bundle publication. The model requires acyclic graphs and SemVer-compatible versions.

a) Shared release logic across operating systems: A game developer can use Windows with Linux dedicated-server CI, or Linux with Windows packaging CI. Shared configuration holds the graph, lifecycle, and portable commands; referenced platform files choose `sh` or PowerShell and override platform-specific scripts. This reuses orchestration logic, without translating shells or making outputs portable. Toolchains must exist on each host. The evaluated conditional wrapper requires POSIX even when a child selects PowerShell; a Windows caller without POSIX must select its configuration directly. Native Windows equivalence and game-development adoption are not measured.

F. A Stable CI Entry Point

The graph drives tests and releases through generic CI. Adding a package changes configuration and scripts; new runners, credentials or tools still require provisioning. Gates reject missing required tests and block failed releases. Affected tasks already appear in Nx [4]; here one package model supports tests and recoverable releases with distinct windows and policies.

a) Observed self-hosting case: CI uses its checkout-built CLI for transitive tests, builds and self-release. This single-project case demonstrates self-hosting, not adoption or causal productivity effects.

III. Formal Model

This section defines the release computation and the publish protocol; Section IV proves guarantees over these definitions. The model separates stage availability from cross-run admission and discharge. It uses one history and per-package versioning; shared-version groups, multi-history composition, VCS adapters, and explicit rollback are outside it. These are forward-release claims; withdrawal activation, inventories and receipts must remain fixed when relating them to the current extended protocol.

A. Stage Availability

For a selected package p, let B_p and P_p denote successful completion of its build and publish stage, respectively. A version task, when present, is a prefix of B_p. For a required provider q and consumer p, the normal successful-stage dependencies include $B_q \to B_p$, $B_p \to P_p$, and $P_q \to P_p$. If q requires published-artifact availability, add $P_q \to B_p$. Here stage arrows mean prerequisite-to-successor; the package edge convention below is consumer-to-provider. The construction is over the full package graph, with stages outside the selected work treated as already satisfied only when their required artifacts are available.

Proposition 1 (stage-order safety)**.** *For a finite acyclic package graph, the constructed stage graph is acyclic. Any schedule that starts a stage only after all its prerequisites succeed respects both local-build and published-artifact availability.*

Proof. Take a provider-first topological rank of packages. Every cross-package stage edge increases that rank. Within a package, the only direction is build to publish. A directed cycle would therefore require a rank decrease or a publish-to-build edge within one package, neither of which exists. Each availability requirement is an explicit prerequisite, so a legal schedule cannot cross it early. The claim concerns ordering,

not whether a command actually produces the promised artifact; that is an integration contract. □

A global build phase followed by a global publish phase violates the $P_{\text{base}} \to B_{\text{app}}$ edge in Fig. 1. Conversely, no edge requires P_{ui} to precede B_{base}, so those stages may overlap. This elementary construction uses standard topological scheduling [14], [15]; the recovery results below address the separate problem of retaining the correct selected work.

For remote local-output consumption, refine $B_q \to B_p$ to $B_q \to X_{q,p} \to B_p$, where $X_{q,p}$ exports, verifies and materializes the exact required outputs. Edge subdivision preserves acyclicity; extra preparation, test and build-only edges require a separate task-DAG check. Local bytes do not replace required publication or durable-recording gates.

B. Workspace, State, Windows

Definition 1 (history)**.** The history is a finite DAG of commits with a distinguished $\mathsf{HEAD} = h$; $reach(c)$ is the set of commits reachable from c by parent edges, including c; write $c' \preceq c$ for $c' \in reach(c)$. h is fixed throughout. Normalization supplies a deterministic total order of the commits consistent with ancestry; "newest" refers to this order, with unit position breaking ties within a commit. Ancestry alone is only a partial order, so the ordering policy is fixed input and cannot use dates. This commit-DAG view underlies Git mining [16] and merge models [17]. Unlike patch-theoretic accounts [18], [19], [20], we model tags and normalized intent, with files and path scopes supplied by normalization.

Definition 2 (workspace)**.** $\mathcal{P}$ is the finite set of packages at h; $E \subseteq \mathcal{P} \times \mathcal{P}$ contains (a, b) when a declares a dependency on b of a propagation kind; $dep(p) = \{d : (d, p) \in E\}$. Let E_o and $E_b \subseteq E_o$ be the publication-order and failure-blocking edges in this model. Optional dependencies may order publication without blocking on failure. The union $E \cup E_o$ is acyclic; cycles are outside this protocol and are rejected before scheduling. For $S \subseteq \mathcal{P}$, $\delta(S, d)$ is the least number of edges on a chain from a member of S to d against the arrows, ∞ if none.

Definition 3 (bumps, versions, channels)**.** $\mathbb{B} = \{\mathsf{none} < \mathsf{patch} < \mathsf{minor} < \mathsf{major}\}$ is a chain, hence a join-semilattice under $\sqcup = \max$. Versions are SemVer 2.0.0 [21] with its precedence; $ch(v)$ is stable for a non-prerelease v and otherwise v's first prerelease identifier; $core(v)$ and $cnt(v)$ are the numeric triple and the prerelease counter; $apply(v, b)$ applies a bump.

Definition 4 (state)**.** A state σ is a finite set of release tags (p, v, c) with $c \preceq h$; states are ordered by inclusion. $base_\sigma(p)$ is the highest-precedence version among p's tags, $sbase_\sigma(p)$ the highest-precedence *non-prerelease* one, $basec_\sigma(p)$ and $scom_\sigma(p)$ their commits; $ch(base_\sigma(p)) = \mathsf{stable}$ by convention when p has no tag. Equal-precedence tags with conflicting release identities are rejected. When no stable tag exists, $sbase$ is the virtual 0.0.0 and $reach(scom) = \emptyset$. An untagged package additionally has $base = 0.0.0$ and $reach(basec) = \emptyset$. Automatic target computation from a virtual stable baseline returns the configured initial version. Within a valid run sequence, tags are never moved or deleted, so states only grow. Rewriting a completion record is outside the guarantees. Tag write access is therefore release authority; a manifest's version field is not the release ledger. Manifests, configuration, and derived scopes remain trusted planning inputs.

Definition 5 (train and fresh-work windows)**.**

$$W_\sigma(P) \;=\; reach(h) \setminus reach(scom_\sigma(P)), \tag{1}$$

and

$$W^\circ_\sigma(P) = W_\sigma(P) \setminus reach(basec_\sigma(P)), \tag{2}$$

An absent baseline contributes empty subtracted reach. The *train window* W_σ retains history since the stable baseline for target computation (Eq. 8); the *fresh-work window* W°_σ governs direct/propagated admission and discharge. Prerelease tags discharge delivered work while retaining train history. A first release covers the whole history, so initial baselines and scope mappings require deliberate configuration.

C. Units, Suppression, Admission

Definition 6 (units)**.** $\mathcal{U}$ is the set of units carried by commits in $\bigcup_P W_\sigma(P)$, parsed and with scopes resolved and corrections applied. For $u \in \mathcal{U}$: c_u is its commit, $scope(u) \subseteq \mathcal{P}$ its resolved scope, $bump(u) \in \mathbb{B}$ the bump its type and `!` map to; its bump axis is $pb(u) \in \mathbb{B}$ (the stated value, or $bump(u)$ under `inherit`; the default once propagation is asked for is `patch` because a dependency update need not change the consumer's own API), $pd(u) \in \mathbb{N} \cup \{\infty\}$ (default 0), $ps(u)$ a scope filter; its channel axis is a channel value $pc(u)$, a depth $pcd(u)$ (default 0) and a filter $pcs(u)$.

Definition 7 (suppression and sources)**.** $canc(c, p)$ holds when c is an ancestor-or-self of a `cancel` commit whose scope contains p; $held(p)$ when p's effective `Release-As` at σ is `none`. Cancellation uses fixed ancestry and scopes; retry claims also fix suppression of outstanding work. With $disch_\sigma(p, c) \equiv c \notin W^\circ_\sigma(p)$,

$$src_\sigma(u) = \{p \in scope(u) : disch_\sigma(p, c_u) \vee \neg(canc(c_u, p) \vee held(p))\}. \tag{3}$$

Suppression affects only *undischarged* work: a later source cancel or hold cannot retract its already-published artifact or strand consumers.

Definition 8 (reach)**.** $R(S, \ell) = \{d \in \mathcal{P} \setminus S : \delta(S, d) \leq \ell\}$. A unit excludes its own sources. Depth is shortest-path distance from that source set; propagated effects never restart propagation.

Definition 9 (channels)**.** $chan_\sigma(P)$, resolved for *every* package: the value proposed by the newest eligible direct `Channel` directive naming P with $c_u \in W^\circ_\sigma(P)$ and $\neg canc(c_u, P)$; else the value proposed for P by the newest eligible channel-axis unit u with $pcd(u) > 0$, $P \in R(src_\sigma(u), pcd(u)) \cap pcs(u)$, $c_u \in W^\circ_\sigma(P)$ and $\neg canc(c_u, P)$;

else $ch(base_\sigma(P))$. A proposal is computed against $cur = ch(base_\sigma(P))$ (the tag, never a value computed this run) and is empty when it proposes cur itself; a transition proposes only where cur matches its `<from>`; a *propagated* non-transition stable never graduates a prerelease package (graduation must be deliberate); `inherit` resolves to the unit's own `Channel` value if any, else to $ch(base_\sigma(q))$ for its sources q, bytewise least name deciding a disagreement. An empty proposal is ineligible; resolution chooses the newest remaining proposal, with direct proposals taking precedence over propagated ones.

Definition 10 (admission and effective bump)**.** Let $live(u)$ mean $bump(u) \neq \mathsf{none}$, $pb(u) \neq \mathsf{none}$, and $pd(u) > 0$. With $srcCh_\sigma(u)$ equal to $\{chan_\sigma(p) : p \in src_\sigma(u)\}$ and

$$rsv_\sigma(u, D) \equiv \mathsf{stable} \in srcCh_\sigma(u) \vee chan_\sigma(D) \in srcCh_\sigma(u) \tag{4}$$

(the modeled policy admits a prerelease source only on a matching consumer channel; this is an explicit release policy, not a universal claim about package installers),

$$\begin{aligned} adm_\sigma(u, D) \;\equiv\; & \\ & D \in R(src_\sigma(u), pd(u)) && \text{(reached)} \\ \wedge\; & D \in ps(u) && \text{(in scope)} \\ \wedge\; & c_u \in W^\circ_\sigma(D) && \text{(fresh window)} \\ \wedge\; & \neg\, canc(c_u, D) && \text{(not discarded)} \\ \wedge\; & live(u) && \text{(bump-live)} \\ \wedge\; & rsv_\sigma(u, D) && \text{(resolvable),} \end{aligned} \tag{5}$$

$$eff_\sigma(P) = dir_\sigma(P) \sqcup \bigsqcup\{\, pb(u) : adm_\sigma(u, P)\,\}, \tag{6}$$

where $dir_\sigma(P) = \bigsqcup\{bump(u) : P \in src_\sigma(u) \cap scope(u),\ c_u \in W^\circ_\sigma(P)\}$. The join cannot lower an existing bump and is independent of gathering order.

The consumer-side staleness audit is the upward dual of Eq. 5, not merely reverse reachability. For candidate D it scans units in $W^\circ_\sigma(D)$, excludes D when it is itself a source of that unit, and applies the same shortest depth, scope, cancellation, *live*, and resolved-channel predicate rsv. These clauses make the upward answer exactly equal to the downward planner's obligations, including prerelease workspaces.

D. Plan, Versions, Runs

Definition 11 (plan and versions)**.** Write $pin_\sigma(P) = v$ when the effective surviving exact `Release-As: v` unit for P has $c_u \in W^\circ_\sigma(P)$; otherwise $pin_\sigma(P)$ is undefined. Thus a successful tag discharges a pin just like other fresh work, and the directive is fixed during a retry.

$$\begin{aligned} \Pi(\sigma) = \{\, P : \neg held(P) \wedge (& eff_\sigma(P) \neq \mathsf{none} \\ \vee\; & chan_\sigma(P) \neq ch(base_\sigma(P)) \\ \vee\; & pin_\sigma(P)\downarrow)\,\}, \end{aligned} \tag{7}$$

each member carrying $\nu_\sigma(P)$. Let $S = sbase_\sigma(P)$, $B = base_\sigma(P)$. Define the train aggregate $train_\sigma(P)$ by Eq. 6, replacing its direct and propagated fresh-window tests $c_u \in W^\circ_\sigma(P)$ with $c_u \in W_\sigma(P)$. All other source, scope, suppression, depth, and resolved-channel predicates stay the same. It computes the target bump over the train, while *eff* determines fresh admission. Write $e = train_\sigma(P)$. First compute the automatic candidate without the pin. An exact pin must exceed B, must not undercut that candidate unless configured leniency explicitly allows it, and must satisfy the configured major-jump limit (default one major above the computed version). An accepted pin supplies the exact version. For automatic versioning, a stable package that remains stable uses $apply(S, eff_\sigma(P))$; a prerelease package graduating to stable uses $apply(S, e)$, preserving the train target. Otherwise, with ch the target channel,

$$\begin{aligned} & t = apply(S, e), \\ & \nu_\sigma(P) = \begin{cases} t\text{-}ch.(cnt(B)+1) & \text{if } core(B) = t \text{ and } ch(B) = ch, \\ t\text{-}ch.0 & \text{otherwise,} \end{cases} \end{aligned} \tag{8}$$

For a channel-only release whose candidate does not exceed B, recompute once with $e = \mathsf{patch}$ (the channel-entry patch); an exact pin is never adjusted this way. This permits entry from a clean stable baseline, for example 1.2.0 to 1.2.1-beta.0, without treating an actual bump regression as a channel switch. The same fallback permits graduation of a train entered only by a channel change. Every computed or pinned version must exceed $base_\sigma(P)$ in precedence, or the run refuses. Recomputing t from the *stable* baseline every run is what moves a whole train to `2.0.0-beta.0` when a breaking change lands mid-train (Table I).

TABLE I
EQ. 8 TRACING ONE TRAIN.

Baseline	Pending bump, channel	Next
`1.2.3`	minor, `beta`	`1.3.0-beta.0`
`1.3.0-beta.0`	no fresh commit	empty plan
`1.3.0-beta.0`	a fresh minor commit	`1.3.0-beta.1`
`1.3.0-beta.1`	minor + a new fix	`1.3.0-beta.2`
`1.3.0-beta.2`	a breaking change lands	`2.0.0-beta.0`
`1.3.0-beta.2`	channel-only, `rc`	`1.3.0-rc.0`
`1.3.0-rc.1`	channel-only, `stable`	`1.3.0`

Definition 12 (run)**.** A run from σ is parameterised by an adversarially chosen set $F \subseteq \Pi(\sigma)$ of packages failing before recorded completion, which induces the blocked set $B = \{p \in \Pi(\sigma) \setminus F : p$ transitively depends on some $q \in F$ via $E_b\}$. Both blocking and publication order traverse all packages, preserving paths through unplanned intermediates. Publication uses $(\mathcal{P}, E_o)$, with bytewise name ties, filtering to the plan *last*. Each package is tagged only after successful publication, including targets that upload nothing (Proposition 2); premature tagging would discharge unpublished work. The abstract transition yields $run(\sigma, F) = \sigma \cup \{(p, \nu_\sigma(p), h) : p \in \Pi(\sigma) \setminus (F \cup B)\}$. The transition abstracts the publish-to-record interval; it does not prove that interval atomic in

the implementation. A *run sequence* is $\sigma_1 \leq \sigma_2 \leq \cdots$ with $\sigma_{k+1} = run(\sigma_k, F_k)$.

The propagation bound covers graph walks, not total planning. Let H and A count reachable commits and parent edges, m the distinct boundary, cancellation and correction/revert markers, and w the machine word width. A child-before-parent pass propagating marker bits costs $O((H + A)\max(1, \lceil m/w \rceil))$ word operations; dense masks use $O(H\lceil m/w \rceil)$ words plus indexes. A commit's bit for marker x is set exactly when it is in $reach(x)$: the marker sets its own bit and each child passes its bits to its parents. Fresh membership needs two boundary lookups from Eq. 2, not one reachability traversal for every stable/fresh pair. Materializing memberships, correction probes, cancellation queries, scope resolution and provenance add work even when their final output is empty. Message scanning is linear in bytes; channel incidence inversion preserves proposal order and complete diagnostics. Walk reuse requires equal source sets, depth and edge kinds; admission and provenance remain per unit. Caches need bounds across all tiers; eviction preserves semantics but can lose the time benefit. Bytewise publication ties cost $O(|E_o| + |\mathcal{P}| \log |\mathcal{P}|)$ with a heap; blocking closure costs $O(|\mathcal{P}| + |E_b|)$ by one multi-source traversal. These indexed phase bounds are informative, not global optimality or distributed speedup guarantees.

Order matters for a reason tags cannot express: a dependent can be published while its promised provider version is unavailable. If both completion tags later point at the same commit, ancestry alone does not reveal that temporal inversion. The protocol therefore prevents it through stage prerequisites. When a publish is rejected because the version already exists (a run died between publish and tag), the protocol requires content-identity verification before adoption and otherwise aborts: tagging a foreign artefact would enrol it into the release lineage. This is a protocol obligation, not a claim that ReleaseWeave can inspect arbitrary user-defined publish scripts; such integrations must supply an idempotency or identity check.

Theorem 1 (determinism)**.** $\Pi(\sigma)$, ν_σ, *the canonical publish order, and the version-guard refusal outcome are functions of the normalized instance I defined below, including its fixed history-ordering policy.*

Proof. Every definition above is a closed expression in those inputs; the three choice points (competing directives, competing propagated channels, publish-order ties) are each resolved by a stated total order: newest commit then last unit within it, and bytewise least name. □

E. Instantiation: What the Theorems Say About a Repository

A normalized instance is

$$I = (\text{history}, h, \mathcal{P}, E, E_o, E_b, \mathcal{U}, \sigma, \textit{config}),$$

Configuration includes history ordering; normalization supplies the inputs. Artifact contents and arbitrary hook effects are abstracted. Guarantees require acyclicity, valid versions and the retry invariant, whose full implementation enforcement is unproved. Section V tests selected behavior; binary refinement remains unproved.

IV. Guarantees and Proof Arguments

Cross-run claims use the *retry invariant*: the corrected tuple stream, source sets, propagation/order/blocking graphs, scope/depth predicates, target baselines, and resolved source/target channel admission of every outstanding unit remain unchanged, except for baseline tags written for successful packages. Two recurring components deserve names. **(H1)** says every baseline channel read by `Propagate-Channel: inherit` remains unchanged; **(H2)** says no contribution initially suppressed by cancellation, hold, or correction is later released at the same h. H1 and H2 are necessary in their respective cases but are not, by themselves, the full retry invariant. Neither is needed for one invocation's determinism.

Lemma 1 (window antitonicity)**.** *For $\sigma' = run(\sigma, F)$, $W^\circ_{\sigma'}(p) = \emptyset$ if p gained its completion tag and $W^\circ_{\sigma'}(p) = W^\circ_\sigma(p)$ otherwise. The train window W also empties after a stable release, but may remain unchanged after a prerelease release.*

Proof. The run adds only tags at h. If p gained none, both baseline commits are unchanged. If it gained a tag, $base_{\sigma'}(p) = h$, so Eq. 2 subtracts $reach(h)$ and is empty. Only a stable release also makes $scom_{\sigma'}(p) = h$. □

Lemma 2 (source stability)**.** *Under the retry invariant, $src_{\sigma_j}(u)$ is constant for every outstanding unit throughout a valid run sequence.*

Proof. The corrected tuple stream and its cancellation, hold, correction, and source-set predicates are fixed components of the invariant. H2 is the component excluding later release of an initially suppressed contribution. Arbitrary tag-set inclusion outside a valid run sequence implies no such monotonicity, because a higher-version tag may point to an earlier commit. □

Lemma 3 (retry stability)**.** *Under the retry invariant, an unreleased package's resolved channel, each outstanding unit's source-channel set, and $rsv(u, d)$ remain constant.*

Proof. These are precisely the channel and source-set components fixed by the retry invariant. Definition 9 then selects the same direct or propagated proposal, and Eq. 4 reads the same sets. □

Remark 1. (H1) cannot be replaced by requiring stable completed releases: `inherit` reads $ch(base_\sigma(q))$, which the origin's own graduation moves: with `core` at baseline `1.5.0-beta.0` and commits `feat(core)^++1: x` and `release(core)%stable: graduate`, run 1 plans the consumer at `2.0.1-beta.0` (inherited `beta`) while `core` graduates; if the consumer fails, run 2 plans it at `2.0.1`: same package, both plans, different versions, every completed

release stable. Thus unchanged inherited-channel state must be an explicit component of the retry invariant; successful stable releases alone do not imply version stability.

Theorem 2 (the eight guarantees)**.** *Write adm_j for adm_{σ_j} and $n = |\Pi(\sigma_1)|$. Under the retry invariant, for every unit u, consumer D and run $j \geq 1$:*

- G1 ***termination.*** *propagation halts after $O(|\mathcal{U}| \cdot (|\mathcal{P}|+|E|))$ graph-traversal work, whatever the declared depths.*
- G2 ***no orphans.*** *$adm_1(u, D) \Rightarrow adm_j(u, D)$ until D releases at a commit containing c_u.*
- G3 ***version stability.*** *if $D \in \Pi(\sigma_1)$ and D is not released before run k, then $D \in \Pi(\sigma_k)$ and $\nu_k(D) = \nu_1(D)$.*
- G4 ***no double ledger delivery.*** *once D releases past c_u, $adm_j(u, D)$ never holds again; with G2, ledger discharge is exactly-once for every contribution whose target eventually succeeds. This does not make external side effects exactly-once.*
- G5 ***no widening.*** *$\{D : adm_j(u, D)\} \subseteq \{D : adm_1(u, D)\}$.*
- G6 ***convergence.*** *$\Pi(\sigma_{n+1}) = \emptyset$, provided each nonempty retry completes and records at least one initially publishable, non-held package.*
- G7 ***channel discharge.*** *a propagated channel contribution is admitted only while its commit remains in $W^\circ(D)$; baseline equality and transition matching independently remove no-op proposals.*
- G8 ***one-way axis independence.*** *bump-axis suppression alone does not suppress a channel change. The converse is false: resolved channels are an input to bump admission in Eq. 4.*

Proof. G1: the traversal computing R marks each package seen at most once, so it does $O(|\mathcal{P}|+|E|)$ work per unit regardless of depth or cycles; acyclicity is needed for the publish order, not here. G2: compare Eq. 5 clause by clause. Source set, depth, graph, scope, cancellation and channel admission are fixed by the retry invariant. The consumer's fresh window is unchanged until D releases (Lemma 1); that release is the only event that can make its admission clause false. G3: D gained no tag, so its baselines, windows and dir are unchanged; the propagated join is unchanged by G2 ($\geq$) and G5 ($\leq$), whose proofs below use only the lemmas; so *eff* is unchanged and the channel is unchanged (Lemma 3). Thus bump-only and channel-only membership remain; an exact `Release-As` and its guard outcome remain the same because the directive, automatic candidate and guard configuration are fixed. In every case ν is the same function of unchanged inputs. G4: by Lemma 1, $W^\circ(D)$ empties after either a stable or prerelease tag, killing fresh direct and propagated admission. This also makes $pin_\sigma(D)$ undefined. This is discharge in the tag ledger; an interruption after an external publish but before its tag requires content-identity reconciliation or safe publisher repetition, not an exactly-once claim about the external operation. G5: every clause of Eq. 5 is constant except the window, which is antitone (Lemma 1); a conjunction of constant and antitone predicates is antitone. G6: $\Pi(\sigma_{j+1}) \subseteq \Pi(\sigma_j)$, since a member of the later plan was not released (else its fresh windows and pins are empty and its channel equals its new baseline) and each disjunct of Π is then unchanged or smaller; by the premise each nonempty retry removes at least one initially publishable package (G4), so $|\Pi|$ strictly decreases to 0 within n runs. The bound assumes progress on every counted nonempty run; it does not bound the number of failed attempts that make no progress. G7: the channel-axis admission test uses $c_u \in W^\circ(D)$, so Lemma 1 discharges it. While fresh, equality with the baseline and a transition's `<from>` check separately reject no-ops. G8: the channel pass runs first and reads no bump-axis output; the bump pass then reads its resolved channels through Eq. 4. □

TABLE II
RUNNING EXAMPLE: PLANNED VERSIONS ASSETS 1.5.0, UI 3.2.1, BASE 0.8.3, APP 2.1.1. THIS MODEL TRACE IS NOT A TIMING MEASUREMENT. NO-PROGRESS ATTEMPTS PRESERVE SAFETY BUT DO NOT COUNT TOWARD THE CONVERGENCE BOUND.

Run	Plan	Recorded success	Failure
1	all four	assets, ui, base	app
2	app 2.1.1	app	none
3	empty	none	none

Proposition 2 (tagging every admitted release)**.** *If a released package with no artefact to upload and a non-empty effective bump were versioned but not tagged, its window would never advance (Lemma 1), it would re-enter every plan forever, and G6 would fail. Hence Definition 12 tags whatever is released, and a fully successful model run has an empty subsequent plan. Held packages are excluded from admission. This necessity is relative to a tag-only ledger; another protocol could use other durable completion records.*

Three consequences explain the operational behavior. *Resumption*: after a partial failure, $\Pi(\sigma_2) = F \cup B$ at the original versions (G2–G5), provided the retry invariant holds. *Idempotence*: let $release(s) = run(s, \emptyset)$ denote a fully successful transformation with fixed reviewed inputs and valid records. Then

$$release(s) = release(release(s)),$$

under the retry invariant, because the first transformation discharges every admitted obligation and the second has an empty plan (G4–G6). Determinism of an unchanged preview is a separate property (Theorem 1). *Scope preservation*: G5 makes the original plan an upper bound on retry publication under the same invariant. This is not an authorization guarantee against changed commands, rewritten tags, or changed credentials.

The idempotence equation concerns recorded release state. If a publisher succeeds but its completion record is lost, its outcome must be reconciled, or the publisher must safely accept repetition. Arbitrary shell-side effects are not made exactly-once by a tag ledger.

V. Evaluation

We ask whether consumer windows retain residual obligations where a source window loses them (RQ1); whether versions and records remain stable through failure, retry, repair, and uncertain responses (RQ2); whether execution respects artifact boundaries and CI gates (RQ3); and what planning costs appear as graph, intent, and history inputs vary (RQ4). The mechanism and companion campaigns use a pinned released Linux ARM64 binary; the historical campaign uses an earlier binary. Exact versions, hashes, environments, and protocols remain in their separate campaign records. None measures the later release-candidate extensions.

TABLE III
Questions, interventions, counting units, and principal outcomes.

RQ	Intervention	Unit	Outcome
1	finite projection; source-window counterfactual	state, transition, episode	0 projection violations; counterfactual loses 260 states and 111 obligations in 84/120 fault episodes
2	faults, repair, identity conflict	140 episodes	140 pass; 10/10 foreign artifacts refused
3	real registries; CI gates	60 executions; 3 CI repetitions	configured barriers and test gates distinguish intended pass/fail paths
4	graph/history factors	8 process cells	.017 s at 100 layered packages; 19.272 s at 10,000

A. Study Design and Evidence Boundaries

The prospective mechanism campaign crosses five four-package DAGs (chain, diamond, fan-out, disconnected pairs, and a chain with shortcuts), stable and beta baselines, four fault targets, and three injection boundaries: before build completion, before publication, and after an artifact is written but the test publisher returns failure. The resulting 120 fault episodes each repeat the fault, repair it, and run once more to test discharge; ten no-fault and ten foreign-identity controls bring the matrix to 140 episodes. The last boundary is an explicit adapter response, not a process crash. Build and publish concurrency are one; seed 20260910 fixes the order.

For RQ1, a finite model projection enumerates ordered four-node DAGs, each single origin, reachable completion sets, and failure subsets. Against the same states and observed episode completions, a computed counterfactual removes a unit when its source completes. This is a mechanism ablation, not a modified binary or competitor. For RQ2, every episode checks the residual plan against the original obligations minus completion tags, preserves target versions, and inspects immutable version/content identity. The local publisher may accept identical content after an uncertain response and must reject foreign content.

Complementary evidence uses a restricted reference model on 300 real Git histories with 5–20 packages, layered DAGs, 4–12 intent units, and mixed tag states; recovery graphs contain 20 or 100 packages. Verdaccio and Docker registries exercise the mixed chains; CI injects failing and missing tests. Historical data cover a broader but imperfect oracle profile, reconstructed policy replay, pinned tool configurations, and eight scale cells. Every campaign retains its own binary, environment, and unit: model states, episodes, histories, fixture executions, and verifier predicates are reported separately. The deterministic matrices do not warrant p-values or a pooled quality score.

B. RQ1: Consumer-Window Mechanism

The finite projection covers 64 ordered four-node DAGs and four single origins: 256 source instances yield 772 reachable completion states, 2,270 failure-subset transitions, and 1,054 progress transitions. It finds no violation of exact residual obligations, no widening, no double ledger delivery, or strict decrease when progress occurs. The source-window mutant loses obligations in 260 of those states. This is an executable stable-channel, one-intent projection, not a proof of the full model.

On binary-derived completion states, the same counterfactual loses 111 obligations in 84 of the 120 fault episodes: 28/40 episodes at each of the build, prepublication, and post-write boundaries, balanced equally between stable and beta. The ten no-fault controls lose none. All ten foreign-identity controls also expose one loss each, reported separately because they test a different intervention. *Answer to RQ1.* Consumer-relative windows retain the expected residual set in the explored states; source completion is insufficient evidence that every consumer has discharged the intent.

C. RQ2: Retry and Identity Stability

All 140 mechanism episodes passed their predeclared outcomes. In the 130 fault or foreign-identity episodes, repeating the adverse condition preserved the plan, target versions, and completed tags; repair produced an empty plan. All 140 final invocations were no-ops with stable tags, and the adapter recorded one artifact write per obligation. Each of the 40 cases at each fault boundary passed, as did all 70 stable and 70 beta cases. The ten foreign artifacts were refused without a new completion tag, after which restoring the intended content allowed recovery. This supports the supplied immutable local adapter, not arbitrary registry publishers or exactly-once shell effects.

Answer to RQ2. Across the designed graph, channel, target, and fault factors, recorded progress survived repeated failure without version drift; identity reconciliation distinguished prior intended writes from conflicting content.

a) Reference agreement and specification regressions.: All 300 fresh core instances produced accepted matching plans, with no differences, refusals, or invalid execution. Of 1,500 historical full-profile histories, 997 produced accepted plans and 503 produced complete E195 refusals for proposed regressive versions. Among accepted plans, 546 comparisons were clean; four refused dossiers also had no row difference, giving 550 matching diagnostic comparisons. Across the profile, 1,914 package rows differed. Rejected diagnostic output is not an executable release plan.

The 1,914 differing rows are retained for diagnosis. They show that the restricted oracle cannot support a full refinement

claim, especially for prerelease and channel behavior. Six historical specification vectors add selected discharge, suppression, channel, and source-exclusion cases without covering the complete language.

b) Earlier recovery and historical replay.: All 20 companion episodes made progress on every run and converged within four runs; the verifier inspected 251 command records. Historical evidence contains two isolated orphan repetitions and 24 broader episodes, which also converged within four runs with zero structural G3/G5 check failures. These checks concern tested schedules and predicates. They do not establish the liveness premise under permanent registry failure, nor do they make an unrecorded external write unambiguous.

Historical fault cells retain orphan and competing-push cases. The latter distinguish completed release tags from a concurrent source change; they test selected interleavings rather than a systematic concurrency sweep.

Five-release historical replay matched 36 of 42 versions in `google-cloud-node` and 7 of 64 in `conventional-changelog`. Those projects did not use the modeled intent language. Their layouts and release policies were reconstructed for the replay, and mismatches may reflect that reconstruction as well as different release conventions. Consequently the experiment demonstrates limits of policy transfer; it is not evidence that our protocol reproduces arbitrary tools' historical version decisions.

D. Historical Comparisons and Negative Evidence

The historical refusal and competing-push fixtures pin Lerna, Nx, and Changesets releases and configurations in their provenance. Lerna and Nx tagged the denied version before its failed upload, while Changesets left it untagged and recovered by querying npm. The observed propagation policies were unconditional for Lerna/Nx, range-gated for Changesets, and per-commit for our protocol. These are observations of the pinned configurations, not universal properties of every mode those tools expose. In competing-push fixtures, compared runs left tags outside branch ancestry; our historical run merged before its final push and planned the new commit. Release-please's tag timing comes from documentation, not an executed comparison [6]. No cross-tool speed or aggregate superiority claim follows.

E. RQ3: Availability Boundaries and CI

The companion Linux ARM64 campaign executes both the three-package npm $\rightarrow$ Docker $\rightarrow$ Docker chain and the four-package running example. Build outputs carry fixture content through OCI archives; private Verdaccio and Docker registries distinguish local output from publication [22], [23]. Each topology repeats five configurations six times in seeded order: mixed and serialized availability, a missing barrier, refused publication, and failed building.

Each topology has 30 initial executions: 12 intended successes and 18 designed failures. Per topology, 12 repairable failures completed with unchanged versions; 24 completed cases then reran without build or publication. Six missing-barrier controls were not repaired. Enabled barriers placed app building after base publication; without one, the app failed against the unavailable image. In all six mixed three-package runs, base building overlapped npm publication.

In the four-package mixed runs, UI building overlapped delayed assets publication. All six base builds finished before UI publication started. This is a valid local-output schedule: base building need not await UI publication, but overlap is not required. Intervals use a shared monotonic clock [24]. An earlier protocol-validation attempt incorrectly required strict overlap and failed its oracle. We corrected the predicate before a full rerun on that earlier binary; the companion campaign reuses the frozen protocol with its own binary pin. A focused oracle test accepts completion before publication starts and rejects starting after publication ends. The original failure is retained separately and is not counted as passing implementation evidence.

The three CI repetitions kept the workflow entry point unchanged while adding packages to shared configuration. They selected required tests through the graph, blocked release stages for an injected or missing test, propagated a specification change to its consumer, and observed a final no-op release rerun. These fixtures support selected ordering, recovery, and CI paths on one Linux ARM64 environment. *Answer to RQ3.* In these fixtures, configured barriers distinguish local build availability from required publication, and both failed and absent required tests prevent release; concurrency and onboarding effort are not measured outcomes.

F. RQ4: Planning Cost

Eight historical cells report seconds and peak MiB. For layered graphs with 200 units, medians rise from .017/15 at 100 packages (185 edges), through .250/21 at 1,000 (2,002 edges), to 19.272/92 at 10,000 (19,919 edges). A 10,000-package chain records 19.354/118; 1,000-package chain and star cells record .254/26 and .244/19. At 1,000 layered packages, 1,000 units give .290/26 and 5,000 padding commits give .664/30. These are contended-host observations, not confidence intervals or a cross-tool benchmark. *Answer to RQ4.* Observed costs increase sharply between the 1,000- and 10,000-package cells; the design does not isolate a general asymptotic or end-to-end bound from history and output costs.

VI. Discussion and Threats to Validity

a) Meaning of the guarantees.: Safety differs from eventual progress [25]. Our retry theorem fixes corrected intent, source membership, admission predicates, and channel inputs. Some stability lemmas follow directly from this invariant, exposing what retries must preserve. New commits, policy edits, or tag rewrites can require replanning; permanent failure need not converge.

The current group profile adds member target floors and independent counter/channel axes. Fixed shared counters can advance on retry and reselect successful members. Its

independent-counter guarantees are a specification contract; neither group mode is proved by our per-package theorem.

The model abstracts artifact contents and combines publication and recording into one transition. Lost responses and failed recording remain integration obligations; Git locks do not make registries transactional. Own-reason failure exceptions, recording, and finalization require refinement beyond the abstract blocked-set rule; selected test traces do not close that gap.

b) Internal and construct validity.: The author developed the system, model, oracle and tests; agreement can retain shared errors. The restricted oracle omits semantics; full-profile categories are heuristic. The prospective matrix was frozen before execution but was not independently replicated. The finite projection covers stable, one-intent states; post-write faults are responses, not crashes. Integrity and concurrency checks cover selected records and interleavings. Histories, rows, episodes, repetitions, commands and assertions remain distinct units.

c) External validity.: Synthetic fixtures establish controlled behavior; historical replay reconstructs policies, and comparisons pin configurations. No user study measures intent correctness or review effort. The control-history polyrepo fits the model; multi-history coordination, worker transport, peer choreography, and their scale remain unevaluated. Platform scripts, agent gates, and ecosystems demonstrate integration scope, not general effectiveness. One contended host cannot establish service-level promises.

d) Exploratory outreach context.: Eight September inventories cover at least 211 deduplicated repositories with unequal review depth; one spans 21 ecosystems. A separate search's 6,731 hits across 1,512 repositories are discovery coverage, not defect counts. RelayCast (9 stars) and Paperclip (80,381) illustrate visibility, not code size. A two-clone RelayCast fixture simulates upload followed by rejected push: no remote tag appears, yet the script reports success. No production failure is claimed. Two observed Astro histories publish npm packages, fail before marketplace publication, then finish green on retry while skipping those destinations; later manual completion was not assessed. An Apache Arrow fixture with a stubbed workflow client selects a stale successful run. These cases motivate destination completion and invocation identity. Healthy, resolved, and later excluded cases remain recorded. This selected review neither estimates defect prevalence nor validates the protocol.

VII. Related Work

a) Release and build systems.: Release engineering [1], monorepo practice [26], and build-system models [15], [27] supply the setting; we claim no new topological or incremental-build algorithm. Lerna, Nx, Changesets, release-please and semantic-release [3], [4], [5], [6], [7] offer versioning/publication workflows; Section V-D compares pinned configurations, not complete tools. Spinnaker's execution histories [28], Nix's input-derived paths [29], TOSCA's lifecycle interfaces [30] and OpenGitOps reconciliation [31] retain their own state models. Our ledger tracks completed releases across independent publication targets.

Versioning studies motivate explicit policy [32], [33], [34], without proving compatibility. EDOS studies installability [35]; CI effects depend on context [36]. DORA associates greater AI adoption with higher delivery throughput and instability [37]. A 2026 qualitative analysis of 1,110 Google engineers' responses collected in 2025 describes verification and production integration costs that can offset AI generation gains [38]. Our skill covers commit release-intent diagnostics, deterministic plan inspection, tests, CI releases and partial-outcome reconciliation. Agent effectiveness and implementation refinement remain unproved. DORA motivates this design, not its validation. Workflow duplication motivates shared configuration [39]; maintenance gains remain unmeasured. Formal VCS models [18], [19], [20], [17] and Git-mining cautions [16] address concerns beyond our release ledger.

b) Recovery and formal assurance.: Buildbox formalizes build, install, ship, and update transitions [40]; Aeolus studies component states, dependencies, and reachability [41]. NEPHELE applies sagas to distributed lifecycle operations [42]; Bui et al. study LLM-assisted CI/CD repair [43]. Sagas [2], transaction limits [44], and independently committed, idempotently processable operations [45], [10] motivate recovery and explicit treatment of uncertain outcomes. We specialize admission and discharge to versioned consumers. Anvil [46] verifies Kubernetes controller implementations; our handwritten model proofs and sampled execution tests provide weaker assurance. Fuzz-evaluation cautions [47] likewise limit generalization from core agreement.

VIII. Conclusion

Stage prerequisites enforce availability; consumer windows retain obligations after provider success. The retry invariant preserves versions and permits convergence. All 140 mechanism episodes retained expected plans; the source-window counterfactual lost work in 84/120 fault episodes. Repository-qualified boundaries and verified worker inputs extend the design; multi-history recovery, recording failures and randomized interleavings need evaluation and mechanized refinement [48].

Data Availability

Retained records include runtime pins, harnesses, raw observations, verified archives, offline verifiers, fixture Git metadata and outreach provenance. They can reveal authorship; no anonymized package is available, and public release remains undecided.

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